

Objective Function

The objective of the sizing and operation problem is to minimize the risk-adjusted annualized life-cycle cost (ALCC). This formulation is simplified from Ljungblom (2025, Eq. 3.1): PV, wind, and SMR generation terms are dropped (grid + BESS only), grid export is removed, and a revenue discount factor captures auction uncertainty for larger DSU bids.

$$\begin{aligned}
 \min Z = & F_{\text{sub}} \\
 & + \underbrace{\text{AF}_{\text{BESS}} (c_P^{\text{cap}} P_{\text{BESS}} + c_E^{\text{cap}} E_{\text{BESS}}) + c_P^{\text{opex}} P_{\text{BESS}} + c_E^{\text{opex}} E_{\text{BESS}}}_{\text{BESS CapEx + O\&M}} \\
 & + \underbrace{\sum_{t \in \mathcal{T}} (\lambda_t + \pi_t) i_t}_{\text{Grid import cost}} \\
 & - \underbrace{\left(1 - \beta \frac{P_{\text{BESS}}}{P_{\text{max}}}\right) R_{\text{DR}}}_{\text{Risk-adjusted DR revenue}}
 \end{aligned} \tag{1}$$

where the demand response revenue R_{DR} is defined as:

$$R_{\text{DR}} = r^{\text{cap}} C_{\text{enr}} \cdot \mathbb{1}(C_{\text{enr}} \geq C_{\text{min}}) \tag{2}$$

Subject to

$$\text{Energy balance: } i_t + d_t = D + c_t \quad \forall t \in \mathcal{T} \tag{3}$$

$$\text{State of charge: } s_t = s_{t-1} + \eta_c c_t - d_t \quad \forall t \in \mathcal{T}, t > 0 \tag{4}$$

$$\text{Initial SoC: } s_0 = S_0 + \eta_c c_0 - d_0 \tag{5}$$

$$\text{Wrap-around: } s_{|\mathcal{T}|-1} \geq S_0 \tag{6}$$

$$\text{Charge bound: } 0 \leq c_t \leq P_{\text{BESS}} \quad \forall t \in \mathcal{T} \tag{7}$$

$$\text{Discharge bound: } 0 \leq d_t \leq P_{\text{BESS}} \quad \forall t \in \mathcal{T} \tag{8}$$

$$\text{SoC bound: } 0 \leq s_t \leq E_{\text{BESS}} \quad \forall t \in \mathcal{T} \tag{9}$$

$$\text{Import bound: } 0 \leq i_t \leq C_{\text{grid}} \quad \forall t \in \mathcal{T} \tag{10}$$

Notation

Symbol	Description	Unit
<i>Decision variables</i>		
i_t	Grid import at hour t	MW
c_t	BESS charge at hour t	MW
d_t	BESS discharge at hour t	MW
s_t	State of charge at hour t	MWh
P_{BESS}	BESS power rating (enumerated)	MW
E_{BESS}	BESS energy capacity = $P_{\text{BESS}} \cdot \tau$	MWh
<i>Parameters</i>		
F_{sub}	Annual grid subscription fee	EUR/yr
AF_{BESS}	Capital recovery factor (annuity factor)	–
$c_P^{\text{cap}}, c_E^{\text{cap}}$	BESS power/energy CapEx	EUR/kW, EUR/kWh
$c_P^{\text{opex}}, c_E^{\text{opex}}$	BESS power/energy OpEx	EUR/kW-yr, EUR/kWh-yr
λ_t	Spot price at hour t	EUR/MWh
π_t	TOU network tariff at hour t	EUR/MWh
D	Constant data center load	MW
C_{grid}	Grid connection capacity	MW
η_c	One-way charge efficiency	–
S_0	Initial state of charge	MWh
τ	BESS duration	h
<i>DR and risk parameters</i>		
r^{cap}	DSU capacity payment rate	EUR/MW/yr
C_{enr}	Enrolled DSU capacity (5-of-10 baseline)	MW
C_{min}	Minimum DSU enrollment threshold	MW
β	Revenue discount factor	–
P_{max}	Maximum BESS power rating considered	MW